Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/28/2006
PI Name:	Jacobs, Christopher Rae Ph.D.		
Project Title:	Loading-Induce Fluid Flow Mediates Microgravity Related Disuse Bone Loss		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	94305-3030	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	12/01/2002	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1601		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The bone loss that occurs with disuse in microgravity is a significant challenge that must be overcome in order for long-duration space flights such as a manned mission to Mars to be successful. Recently it has been shown that interstitial fluid flow in bone due to loading is a potent physical signal in regulating bone cell metabolism. The purpose of this project is to determine if this physical signal is also important in the formation of new bone forming cells (osteoblasts). To determine if bone fluid flow due to loading can regulate the formation of new bone cells in terms of proliferation and differentiation of osteoblast precursor cells. We will also determine if the biochemical mechanism involved is similar to the one we have desribed involved in the regulation of mature bone cells. An understanding of the cellular and biochemical mechanism involved in the regulation of bone metabolism due to mechanical loading will result in novel targets for pharmacological intervention, potentially leading to a new class of drugs which could mimic the effects of loading on bone, even in the absence of loading or in the presence of reduced loading.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Disuse bone loss is a critical problem in settings other than microgravity. These include osteoporosis related to age, since inactivity is a known contributing factor, as well as bone loss that occurs with paralysis and stroke.
Task Progress:	The bone loss that occurs with disuse in microgravity is a significant challenge that must be overcome in order for long-duration space flights such as a manned mission to Mars to be successful. Recently it has been shown that interstitial fluid flow in bone due to loading is a potent physical signal in regulating bone cell metabolism. The purpose of this project is to determine if this physical signal is also important in the formation on new bone forming cells (osteoblasts). Thus, the objective of this project is to determine if bone fluid flow due to loading can regulate the formation of new bone cells in terms of proliferation and differentiation of osteoblast precursor cells. The study will be conducted with bone marrow stromal cells that contain osteoblast precursors and our in vitro cell culture flow system. Outcome variables will include proliferation rate determined by BrdU incorporation, PGE2 release, and osteogenic gene expression. We will also determine if the biochemical mechanism involved is similar to the one we have desribed involved in the regulation Significance: An understanding of the cellular and biochemical mechanism involved in the regulation of bone metabolism due to mechanical loading will result in novel targets for pharmacological intervention potentially leading to a new class of drugs which could mimic the effects of loading on bone even in the absence of loading or in the presence of reduced loading. Our results will also have important terrestrial application. Disuse bone loss is a critical problem in settings other than microgravity. These include osteoporosis related to age since inactivity is a known contributing factor as well as bone loss that occurs with paralysis and stroke. In the last year of the project we have completed the majority of the data collection phase of the project. Specifically, we have demonstrated that fluid flow due to loading can increase the proliferation of osteoblast precursors as well as increasing their differentiation into mature osteoblasts. We have also made a related discovery that the dynamics of the timepattern of loading can have a significant effect on the impact of fluid flow on bone cell metabolism. We are currently working to better define the time course of the effect of flow on gene expression in osteoblast precursors. We are currently analyzing the results for the last phase of the project to relate the signaling pathway activated by flow in precursors to that found in mature osteoblasts. In order to complete this analysis and publish our final findings we anticipate requesting a one year no-cost extension for the project.
Bibliography Type:	Description: (Last Updated: 02/08/2013)
Articles in Peer-reviewed Journals	Li YJ, Batra NN, You L, Meier SC, Coe IA, Yellowley CE, Jacobs CR. "Oscillatory fluid flow affects human marrow stromal cell proliferation and differentiation." J Orthop Res. 2004 Nov;22(6):1283-9. PMID: 15475210 , Nov-2004
Articles in Peer-reviewed Journals	Batra NN, Li YJ, Yellowley CE, You L, Malone AM, Kim CH, Jacobs CR. "Effects of short-term recovery periods on fluid-induced signaling in osteoblastic cells." J Biomech. 2005 Sep;38(9):1909-17. PMID: 16023480 , Sep-2005